Time dilation from counting events without pre-assigned places, and the Feynman checkerboard up to a moving atomic clock: registered lattice tests

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Version 2.40.2, 2026-10-25. Archived on Zenodo, https://doi.org/10.5281/zenodo.23631239. Licence CC BY 4.0. Developed with the help of AI assistants, as declared in the record.

Abstract

This record reports a sequence of lattice tests motivated by four counting premises and implemented through known physical constructions, with the further assumptions stated explicitly. With a light-cone order and a clock that reads the longest chain of events, independent uniform random events recover proper time within the reported finite-size accuracy, while a synchronous grid gives a chain length that does not depend on the velocity; a clock that counts events and multiplies the two light directions recovers the Lorentz factor on that same grid, through the exact identity $\sqrt{(ab)}/((a + b)/2) = \sqrt{(1 - v^2)}$. With an imaginary weight for every change of light direction the count becomes the Feynman checkerboard, a Dirac quantum walk; the subsequent tests concern mass as the turning angle, fermion doubling with net chirality zero, boundary (domain-wall) fermions, the gauge field of MM 2.27 on the same lattice (Landau levels, Maxwell's evolution equations with Coulomb's law in 2D and 3D, photons with two polarizations, a fitted Casimir energy coefficient of 0.9997 times $-\pi^2/720$), the Schwinger model in 1+1, and hydrogen-like atoms: for a Bohr radius of six lattice steps the emission rates agree within 1.7 % with the nonrelativistic hydrogenic formula at the same coupling and effective mass.

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Version 2.40.2 (25 October 2026). This version adds section 3.1: with more than two counted sides, the product of all sides is not Lorentzian, the product taken pair by pair is, and n sides give n − 1 spatial directions, so 3+1 corresponds to four sides. It adds to section 13 the precedent of Knuth and Bahreyni (2014), not cited before, and a 2016 report by Knuth on the number of chains, and it states in section 12 that the number of spatial dimensions is not derived. Everything of version 2.40.1 (programs, outputs, registered tests and their totals) is unchanged. Programs, outputs and registrations of the addition are in `aggiunta_2.40.2/` (ADDITION.md).

Version 2.40.1 (15 October 2026). This version corrects two statements of version 2.40 (10 October 2026). The massive palindromic automaton has 64 Dirac valleys, not 16 species (section 5). With the parameters used, the 3D hand on the boundary also has boundary branches near half period (section 6). The original programs, outputs and registered-prediction totals are unchanged; this version corrects the explanatory text and adds supplementary checks. Details, programs and the registration are in `correzione_2.40.1/CORRECTION.md`.

MM is the author's framework, in which physical quantities are read off from counts. This deposit uses only the small part of it stated in section 2. Ugo Lanciano is not a professional physicist; the computations were done with an AI assistant (section 16).

In brief. With a light-cone order and a clock that reads the longest chain of events, independent random events recover special-relativistic proper time, while a synchronous grid gives a chain length that does not depend on the velocity; a clock that counts events and multiplies the two light directions recovers the Lorentz factor on that same grid. Weighting every change of light direction by an imaginary amount turns the count into the Feynman checkerboard, a Dirac quantum walk. On that lattice we add the gauge field of MM 2.27 and test, one step at a time, Landau levels, Maxwell's evolution equations, photons, the Casimir energy and a hydrogen-like atom, whose emission rates agree within 1.7 % with the hydrogenic formula at the same coupling. For a moving atom, the lattice computations at the spacing we could afford do not show Einstein's slowdown of its internal frequency; with the free dispersions replaced by the continuum ones the coefficient becomes −0.44 (−0.49 in the strictly first-order ratio) against −0.50. All the structures are known physics; the deposit assembles them as a sequence motivated by four counting premises, with the further assumptions stated. Every prediction was hashed before computing, and those that failed are reported with the others. With more than two counted sides, only the product taken pair by pair keeps the Lorentz factor (section 3.1).

1. What is known, and what is proposed

Every result below has a known counterpart: the longest chain of a Poisson sprinkling as proper time (causal sets), the Feynman checkerboard and the Dirac quantum walks, Wilson and domain-wall fermions, lattice electrodynamics, the Schwinger model, the golden rule for spontaneous emission, the Breit interaction. Section 13 lists the sources and what differs here.

What this deposit proposes is a single sequence that starts from counts and reaches a moving atomic clock:

1. a clock that counts events without pre-assigned places runs at the geometric mean of the two light sides, which is Einstein's proper time; a clock that reads the longest chain on a grid of places gives the arithmetic mean, which is Galilean time (section 3); with more than two counted sides only the pairwise product keeps the Lorentz factor, and n sides give n − 1 spatial directions (section 3.1, added in 2.40.2);
2. the same count with an imaginary weight per turn is the checkerboard: a Dirac quantum walk, with spin ½ required in the three-dimensional construction used here, and mass as the turning angle (sections 4-5);
3. massless one-handed fermions live on the boundary between two regions of opposite turning, and two such boundaries bind into a mass that falls with their distance (section 6);
4. the field of MM 2.27, put on the same lattice, gives gauge invariance, Maxwell's evolution equations, photons and an atom that emits light (sections 7-9);
5. the atom in motion: on the lattice computed here its frequency does not slow down; with continuum dispersions it does, close to Einstein's coefficient (section 10).

The sequence is tested, not derived as a theorem. The four premises of section 2 motivate it but do not determine it alone: the light-cone order, the Dirac matrices and the order of the steps, the Wilson term, an extra direction for the boundaries, the field equations of MM 2.27, canonical quantization, the forms of the interactions and the no-pair projection are further assumptions, stated where used. Each step is a numbered test with predictions registered beforehand (`documenti/REGISTERED_TESTS.md`; originals in `documenti/registrations_original/`).

2. The model: four premises and the declared choices

Units: lattice step a = 1, c = 1, ħ = 1. Light moves one step per tick; a light-cone order is assumed.

Premises of the framework, in technical form:

- **A (no pre-assigned places).** Events ("copies") are not placed on a set of places given in advance. In the tests they are taken as independent and uniformly random in spacetime volume; this is an assumption added to A, which A motivates but does not imply.
- **B (second link).** In an elementary step the two light-side counts a and b of a clock are coprime. Copies are generated at their common multiples, so a light diamond of sides a and b holds ab = gcd(a, b)·lcm(a, b) cells.
- **C (opposite directions multiply).** A clock reads the number of copies and combines the two opposite light directions by product: its time is √(ab).
- **D (histories that never happened).** Each change of light direction of a history (a "turn") carries an imaginary weight; the amplitude of an outcome is the sum over all histories.

The gauge field is the one of MM 2.27 (phases on the links, the equations of its script F, α = β = 1).

Declared choices, each stated where used:

- **Wilson term** r: the turn depends on the neighbours, so that the extra species of the cubic lattice become heavy (section 5; 63 extra Dirac valleys of the massive automaton, corrected in 2.40.1);
- **palindromic order** of the axes within a tick, which removes a spurious spin splitting (section 4);
- the regularised Coulomb potential $-Z\alpha/\sqrt{(r^2 + 0.25)}$;
- the **continuous-time form** of the checkerboard, the lattice Dirac Hamiltonian $H = \Sigma_a \alpha_a \sin p_a + \beta(m + r \Sigma_a(1 - \cos p_a))$, used for the 1+1 electrodynamics (section 8) and for the atoms (sections 9-10). Sections 4-7 use the discrete-time automaton itself; section 3 only counts events.

What the framework does not fix: the value of any single mass (the turning angle θ is a free parameter, as m is in Dirac's equation), and the lattice step in physical units (if it is the Planck length, the electron accumulates one radian of mass phase in about 2.4×10^{22} ticks).

3. The clock (P1-P7)

Between the start and end events of a clock moving at speed v, light rays bound a diamond with sides a = Δt − Δx and b = Δt + Δx.

arrangement of copies	what the clock reads	result
uniform at random, 1+1, 200 000 copies	longest chain	$\sqrt{(1 - v^2)}$ within +0.2 to −0.9 % for v = 0.3-0.9 (within 0.3 % after a finite-size correction)
uniform at random, 3+1, about 6000 copies	longest chain	moving and resting clocks with the same proper time agree within 1.3 σ; the fixed tolerance of 6 % failed at v = 0.8 (about 780 copies)
synchronous grid	longest chain	(a + b)/2 for every v: a velocity-independent chain length
fixed places, random times	longest chain	dilation 7-29 % stronger than Lorentz: the places define a frame
synchronous grid	number of copies, product rule (B, C)	$\sqrt{(1 - v^2)}$ within 0.43 %
light lattice	√(ab) against (a + b)/2	$\sqrt{(1 - v^2)}$ to 3×10^{-16}

The number of copies in the diamond is proportional to ab; a clock that reads √(ab) runs at the geometric mean of the two sides, while a sum of the sides gives the arithmetic mean. **The arithmetic mean exceeds the geometric mean except at rest, and the gap is time dilation:** $\sqrt{(ab)}/((a + b)/2) = \sqrt{(1 - v^2)}$, an exact identity. A clock at rest steps by (1, 1); a step with coprime sides is indivisible (the segment from (0, 0) to (a, b) has gcd(a, b) − 1 interior lattice points). With unequal sides in every step, a clock at rest would zigzag and tick like a straight clock at c/3 (P7c): this is why the resting step is (1, 1).

The success criterion registered for P1-P5 as a whole ("if P1 or P5 fails, the right angle does not follow from the count") was not met, because P5 failed its 6 % tolerance at v = 0.8; the equal-proper-time comparison of P5 passed, and the finite-size reading is an interpretation made afterwards. The same tests, with the same programs, are section 10 of MM 2.36.1, where they are called M1-M7 and premises A and B are called P7 and P6.

3.1 More than two sides: the pairwise rule (added in 2.40.2)

Premise C multiplies the two light sides of a clock. If a clock is described by n independent counts $x_1, \ldots, x_n$ (its sides; a and b for n = 2), premise C can be extended in two ways, which coincide for n = 2: the **product rule**, a clock reading the geometric mean $(x_1 \cdots x_n)^{(1/n)}$, with an order in which every count increases; and the **pairwise rule**, in which each pair of sides multiplies and the pairs add, $Q = \Sigma_{i<j} x_i x_j$, with a clock reading $\sqrt{Q}$ and an order in which the total $T = \Sigma x_i$ increases and Q of the difference is positive.

- Q has signature (1, n − 1) and equals $((n - 1)/(2n))T^2 - \tfrac{1}{2}\Sigma(x_i - \bar{x})^2$: one time direction, the common growth of the sides, and n − 1 space directions, their spread, in which the form is isotropic (the permutations of the sides are isometries). Three sides give 2+1, four give 3+1; for n = 2, Q = ab. This holds for every n (it follows from the Gram matrix of the sides).
- For n ≥ 3 the only diagonal map with positive factors that preserves Q is the identity ($\lambda_i\lambda_j = 1$ for every pair forces $\lambda_i = 1$), while the product, preserved by rescalings with $\prod\lambda_i = 1$ and by permutations but not by rotations, has no Lorentz group.
- On structures built from independent rank counts, without an embedding in Minkowski coordinates, four sides with the pairwise rule give a Myrheim-Meyer dimension of 3.92 ± 0.11 on the six sampled intervals estimated to lie at least 95 % inside the region (3.61 on all sampled intervals), and first-observer moving-clock ratios of 0.981 and 0.998 (means over 10 seeds in the classes v = 0.2-0.4 and 0.4-0.6); three sides give 2.98 ± 0.05 and 0.977, 0.994. The same four counts with the product rule give a structure of dimension about 3.7 whose moving clocks are 12 % off (1.121). A control with Poisson copies in 3+1, measured by the same code, gives 0.992 and 1.024.

The algebra is known: Q is the Minkowski interval along n null directions with equal mutual products (section 13). The pairwise rule defines, for every n, a Lorentzian form with n − 1 spatial directions; the tests on rank structures cover three and four sides, and there the pairwise rule survives while the product rule does not. The number of sides is not derived (section 12). Limits: the registered two-observer comparison of the moving-clock test failed for both pairwise structures (−3.4 % and −7.5 % in the faster class), and that form of the test is not validated beyond 1+1, so only the first observer is used above; the estimator reads 3.61 on all sampled intervals, because the region of the ranks is not causally convex for this order (a registered diagnostic separates the intervals estimated to lie inside it); the registered prediction for the

longest-chain dimension also failed (3.45), and that diagnostic does not address it; the isotropy of the sampled structures was not tested, only that of the form. Programs, outputs, registrations and the count (20 passed, 4 failed, 4 not evaluable or incomplete) are in `aggiunta_2.40.2/`.

4. The checkerboard (P8-P9)

In 1+1, a history steps along one of the two light directions at each tick, and every turn carries an imaginary weight of size $\theta = \varepsilon m$ (premise D; the registration writes it $+i\varepsilon m$, the code $-i\theta$: a sign convention). Summing all histories (Feynman and Hibbs 1965, problem 2-6):

- the normalized amplitude A converges to $J_0(m\tau)$, with error 3.5×10^{-3} at $\varepsilon m = 0.002$, halved (ratio 2.01) when the step is halved; the other component carries the factor $\sqrt{((1 + v)/(1 - v))}$ in front of J_1, as a spinor component should;
- A depends on the proper time τ only, up to $v = 0.95$, within 7.7×10^{-3} (Lorentz invariance in the limit of fine steps);
- the first zero of the moving clock is delayed by γ within 0.06 %;
- with real weights instead (the classical zigzag of Kac), the sum depends on the laboratory time: 0.4607 against $e^{-a\tau(\gamma - 1)} = 0.4614$. The imaginary weight is what makes the count Lorentz invariant.

In 3+1 the history takes one step along x, y and z at each tick, each in one of the two light directions of that axis, chosen by the spin; then the turn exchanges the chiralities. The four matrices anticommute, so the squares add without cross terms:

- $\omega^2 = k_x^2 + k_y^2 + k_z^2 + m^2$ within 1.4×10^{-3}; group velocity within 3.8×10^{-4} and parallel to k; clock rate $m\sqrt{(1 - v^2)}$ within 9.4×10^{-4}; error divided by 4.00 when the step is halved;
- without the palindromic order the error of the dispersion rises to 4.3 % on the face diagonal at 0.9 c (estimated spin splitting 4.26 %);
- without spin, a specific scalar zigzag on six directions has three modes at rest that do not propagate: **in this construction spin ½ is required.**

This is a split-step member of the family of Dirac quantum walks (Bialynicki-Birula 1994; Meyer 1996; Arrighi, Nesme and Forets 2014; D'Ariano and Perinotti 2014).

5. Mass, and relations among particles (P10-P11)

- **Mass is the turning angle per tick:** the rest frequency equals θ exactly (error $\leq 2 \times 10^{-16}$).
- **In the one-dimensional automaton the band becomes flat at $\theta = \pi/2$** ($|\partial\omega/\partial k| \leq 6 \times 10^{-14}$): a turn at every tick and a particle that cannot move. This is a property of that automaton ($\cos\omega = \cos\theta \cos k$), not a general bound: in the palindromic 3D automaton the band at $\theta = \pi/2$ is not flat.
- **Rest and inertial mass** coincide only for small θ: inertial/rest $- 1 = \theta^2/3$ in 1D (3.36×10^{-5} against 3.33×10^{-5} at $\theta = 0.01$, in a recheck with a finer derivative step) and about $4\theta^2/3$ in the 3D automaton ($1.34\ \theta^2$).
- **Particle and antiparticle** have the same mass (spectra symmetric to 2×10^{-15}).
- **Free fall,** when a potential stretches the local step as the lapse of MM 2.36 does: $a = -g\theta/\tan\theta$ within 0.12 %. Light particles ($\theta \ll 1$) fall alike; the finite-step correction, $(\theta_1^2 - \theta_2^2)/3 \approx (m/m_P)^2/3$ between two masses, depends on the mass.
- **Hands come in pairs:** the massless (Weyl) automaton has 16 nodes, 8 at $\{0, \pi\}^3$ with chirality +1 and 8 at $\{\pi/2, 3\pi/2\}^3$ with chirality −1, net zero; no small local turn opens a gap (10 cases out of 10). The massive (palindromic, two-tick) automaton has 64 Dirac valleys with gap θ in the Brillouin cube $[0, 2\pi)^3$, 32 around quasi-energy 0 and 32 around $\pi/2$, that is 8 in each of the eight parity sectors conserved by the two-tick evolution without the Wilson term (corrected in 2.40.1: version 2.40 said 16 species, having checked 16 of the 64). This is consistent with fermion doubling and with the Nielsen-Ninomiya theorem, whose hypotheses concern static Hamiltonians.

The framework does not fix the value of θ for any particle; it fixes the relations above.

6. The hand on the boundary (P12-P15)

With a turn of opposite sign in two regions, $\theta(s) = \pm\theta_0$ ($\theta_0 = 0.2$), and the Wilson term, one-handed massless states live on the boundaries:

- 2D strip, r = 0.3: one mode per boundary, slopes +1.00000 and −1.00000, $\langle\sigma_x\rangle = \pm 1.0000$ (in a recheck with a k

grid shifted off k = 0; the registered detector failed there); without the Wilson term, four right-handed and four left-handed modes per boundary;
- 3D (a fourth direction s across the boundaries), r = 0.15: near quasi-energy 0, two states per boundary with $\omega = \pm|k|$ ($\omega/|k| = 1.0000$), helicity ±1.000, chirality +1 on one boundary and −1 on the other; near quasi-energy 0, no boundary states in the sampled neighbourhoods of the other 15 special points; with r = 0, 60 extra states per boundary;
- near half period (quasi-energy π/2 per tick; added in 2.40.1) there are additional boundary branches, which we call temporal doublers of the hand on the boundary: on a sampled 24^3 grid, light in-gap states occur near π/2 only at k = 0 (8 states) and at the six points $(\pm\pi/2)e_j$ (4 each). They are consistent with a mass-sign criterion: at these valleys the Wilson mass (r) is smaller than θ_0, so the effective mass changes sign across the boundary. Their mean distance from π/2 at the separation used here is 0.0127 at k = 0 (0.0122 at $(\pi/2)e_x$) and decreases approximately exponentially with the separation, at a rate close to 0.05 per site. In a control with $\theta_0 = 0.1 < r$ they are absent (`correzione_2.40.1/CORRECTION.md`);
- two boundaries at distance d bind into a particle with a true mass ($\omega^2 - k^2 = m^2$ within 3.5 %), shared half and half between the boundaries; the mass falls nearly exponentially with d: m/θ_0 from 1.0×10^{-2} at d = 10 to 1.9×10^{-4} at d = 35 in 3D (local rate 0.13-0.18 per site, growing with d, the same as in 2D within 1.4 %).

A small mass does not need a small number: it needs distance. The localization goes back to Jackiw and Rebbi (1976); the chiral lattice construction is the domain-wall fermion of Kaplan (1992) and Shamir (1993), here inside the checkerboard.

7. The field of MM 2.27 on the checkerboard (P16-P19)

- **Gauge field as phases on the links:** the spectrum is invariant under local phase changes (1.5×10^{-14}); in a magnetic field the levels are $\pm\sqrt{(m^2 + 2nB)}$ within −1.0 to +1.6 % for n = 1-3, with the n = 0 level on one side only. In an electric field, over the tested range $p \le 3\theta$, the speed stays below 1 (maximum 0.945) and approaches the continuum value $v = p/\sqrt{(p^2 + \theta^2)}$, with oscillations at switch-on (Zitterbewegung, period about 32 ticks) that made the registered 2 % tolerance fail.
- **Dynamic field, 2D:** the evolution equations of MM 2.27 (φ on nodes, A on links, leapfrog with dt = 1/2), sourced by the checkerboard current; the coupling is staggered by half a tick, so Gauss's law is not exact, as already in MM 2.27. Charge conserved to 8.7×10^{-18}; the field of a charge reaches distance 24 with its maximum at t = 28; like charges repel and opposite charges attract with nearly equal and opposite displacements (ratio 1.037 on a 192 × 192 torus, in a recheck after the registered 96 × 96 torus proved too small).
- **Dynamic field, 3D:** charge conserved to 3.5×10^{-18}; at R = 20 the field reaches 1 % at t = 11 and peaks at t = 21; Coulomb's law $E \cdot 4\pi r^2/Q = 0.988, 0.990, 0.970, 1.077$ at r = 10, 14, 18, 22; attraction and repulsion symmetric (ratio 1.001).
- **Quantized free field, 3D:** two polarizations (rank 2 to 1.4×10^{-14}), $\omega/|k|$ within 0.36 % for $|k| \le 0.3$; black-body energy within 2.5 % of Stefan-Boltzmann at T = 0.1; **the fitted Casimir energy coefficient is 0.9997 of $-\pi^2/720$** (energy per area C/d^3; the force follows as $3C/d^4$); zero-point energy 2.39 per cell.

These are numerical checks of the evolution equations adopted, not of the full set of Maxwell's constraints.

8. Quantized field and matter, 1+1 (P20)

Exact diagonalization (up to 184 756 states) of the Schwinger model on a lattice, the flux on the links quantized by Gauss's law and the matter on staggered sites (Kogut and Susskind 1975):

- confinement: the energy between two static charges grows as $g^2/2$ per link (slope 0.4974 against 0.5);
- string breaking: with light matter the vacuum creates a pair that screens the charges (slope 0.290 → 0.022; induced charge −0.993);
- the Schwinger mass is not reached at these sizes: 0.732 against $1/\sqrt{\pi} = 0.564$ at N = 20, decreasing slowly with N (0.760, 0.744, 0.732 for N = 16, 18, 20). The continuum limit is not established at these chain lengths.

9. An atom that emits light (P21-P22)

The electron is the continuous-time checkerboard with r = 0.2, the nucleus a fixed charge, the photons the quantized modes of section 7, coupled through the link phases with the physical $\alpha = 1/137.036$; emission rates from the golden rule at leading order.

- **Small atom** (64^3 torus, m = 0.8, $Z\alpha = 0.6$, Bohr radius 2.4 steps): 1s, 2s and 2p exist, with the fine structure in Dirac's order, and the lattice coupling equals the dipole form in the long-wavelength limit (0.999999). All four registered predictions failed: in the registered run the 2p-1s frequency is 75 % above the continuum and the rates

are 34-44 % below the hydrogenic formula (−32 % and −46 % after re-diagonalizing H in the odd sector, a recheck). The atom is too small: the 1s feels the lattice, and the photon wavelength is comparable to the atom.
- **Large atom** (144^3 torus, 11.9 million components, m = 0.625, Zα = 0.3, Bohr radius 6.0 steps): all five predictions passed.

quantity	$2p_{1/2} \to 1s$	$2p_{3/2} \to 1s$
frequency against $(3/8)\, m^*(Z\alpha)^2$	+1.04 %	+1.66 %
golden-rule rate against the dipole formula	+0.83 %	−3.68 %
rate against the hydrogenic formula $(2/3)^8\, \alpha\, (Z\alpha)^4\, m^*$	+0.99 %	−1.67 %
identity between the length and velocity forms	0.999936	0.999933

Spectrum: 1s 0.5996603, 2s 0.618515, $2p_{1/2}$ 0.6186051, $2p_{3/2}$ 0.6187224. $m^* = m/(1 + rm)$ is the kinetic mass of the Wilson electron; the comparison is with the nonrelativistic hydrogenic formula at the same Zα and m*, not with physical hydrogen.

10. The moving atomic clock (P23-P25c)

The observable. The clock is the energy gap, at fixed total momentum P, between the centroid of the eight n = 2 states (2s and 2p) and the 1s doublet; at rest the 2p-1s gap is also reported. $R = \omega(P)/\omega(0)$. The velocity in the tables is nominal: $V = P/\sqrt{(m + M)^2 + P^2}$, with the free masses. For a relativistic composite, $\omega(P) = \omega(0)/\gamma$, that is $R - 1 = -V^2/2$. The frequency of an emitted photon, with recoil and direction, is not computed.

The model. Electron and nucleus (a scalar particle with the lattice dispersion $\sqrt{(M^2 + 4\Sigma \sin^2(q/2))}$), reduced to the relative coordinate at fixed P; m = 0.625, M = 2.2222, Zα = 0.375, 144^3 grid of relative momenta (P is not a multiple of 2π/144, so the torus carries twisted boundary conditions; equivalently, the infinite-volume problem sampled on that grid). Interactions: instantaneous Coulomb, and a Breit operator added at first order: the regularized continuum form $V_B = (Z\alpha/2)\, \Sigma_{ab}\, \alpha_a\, T_{ab}\, v_{N,b}$ (symmetrized), with $T_{ab} = \delta_{ab}/r_s + r_a\, r_b/r_s^3$, $r_s = \sqrt{(r^2 + 0.25)}$, the electron Dirac matrix α and the nuclear velocity v_N from its dispersion. A derivation of the magnetic exchange from the Wilson lattice current has not been made: for the Wilson electron the free velocity is $\alpha_a \cos k_a + \beta r \sin k_a$, not α.

The continuum expectation. For the leading Coulomb binding, at first order in the relativistic corrections and at second order in the center-of-mass velocity, averaged over the multiplet: the kinetic terms give $R - 1 = +(5/6)V^2$ and the Darwin-Breit term $-(4/3)V^2$, total $-V^2/2$ (the Breit term is what makes the binding energy contribute to inertia; Krajcik and Foldy 1974). It is not exact at Zα = 0.375 with a regularized potential and a finite grid.

The ratio R_B is formed from energies corrected at first order; expanding the ratio itself strictly to first order in the Breit shift gives a second value, also listed.

test	dispersions	$(R - 1)/V^2$, Coulomb only	Breit contribution	$(R - 1)/V^2$, with Breit	same, strictly first order
P23-P24	Wilson r = 0.2	+1.125 (V 0.15), +1.164 (V 0.30)	−1.116, −1.079	+0.010, +0.085	−0.037, +0.040
P25b	Wilson r = 0.2 / 0.1 / 0.05	+1.134 / +1.188 / +1.215	−1.131 / −1.161 / −1.176	+0.003 / +0.027 / +0.039	−0.042 / −0.023 / −0.014
P25c	continuum, on the lattice momenta	+0.778	−1.215	−0.438	−0.488
continuum, leading order	—	+0.833	−1.333	−0.500	−0.500

At rest the atom is right: frequency 0.984 and binding 0.990 of the reduced-mass values (P23). In motion, on the Wilson lattice, the Breit term has nearly the expected size but the kinetic part is too large, and the coefficient stays near zero, its sign depending on the convention: far from Einstein's −0.5. The registered prediction that reducing the Wilson term would restore the slowdown (P25b) failed: the coefficient does not move towards −0.5 as r decreases. This does not

show that the Wilson term plays no role.

What P25c shows. On the lattice the nucleus has M = 2.2, a Compton wavelength shorter than one step, and moving the atom at V = 0.15 takes a total momentum of 0.43 per step; there the lattice dispersions differ from the relativistic ones by terms of the same order as the effect. With the continuum free dispersions on the same momenta (electron $\alpha \cdot k + \beta m$, nucleus $\sqrt{(M^2 + q^2)}$), the same Coulomb, the same Breit operator and the same no-pair method, the coefficient is −0.438, or −0.488 strictly at first order, against −0.500. This change also makes the electron kinematics consistent with the vertex α used in Breit. P25c therefore supports the reading that coarse lattice kinematics contributes substantially to the missing slowdown. It is not a direct test of the interacting continuum limit of the checkerboard: that would also require the current and the magnetic interaction derived on the lattice, control of the no-pair approximation, of finite-volume and cutoff effects and of perturbative errors, with electron and nuclear masses and momenta all small in lattice units. Such convergence has not been demonstrated here. The residual difference (12 % in the unexpanded ratio) is of the order of the other lattice effects at this size, as an order-of-magnitude statement, not an error budget.

Two technical points.

- In P25 (Wilson r = 0.1 and 0.05) and in one run of P23 (lighter nucleus), the free band of states with a negative-energy electron and a fast nucleus, $-E_e(k) + \varepsilon_N(P - k) - M$, reaches 0.67 at r = 0.1 and 0.97 at r = 0.05, above the 1s (0.59); for r = 0.2 and M = 2.2222 its top is 0.07. The eigensolver on $(H - \sigma)^2$ did not converge there (its residual, of the squared operator, stayed near 10^{-5}, with six approximate energies within 2×10^{-4}). P25 is not concluded.
- P25b and P25c use the no-pair projection with the free projectors $\Lambda_+(k) = (1 + h(k)/E(k))/2$ (Sucher 1980), with the negative-energy subspace shifted above the spectrum. This is a change of the model, not only of the solver: it drops the Coulomb couplings to the negative-energy states. At r = 0.2, where both methods work, it agrees with P23 within 0.65 % in frequency and within 0.01 in $(R - 1)/V^2$, a numerical check, not an identity. The residuals and the negative-energy part ($\leq 2.4 \times 10^{-14}$) were checked on the eigenvectors before they were saved in single precision for the Breit step.

11. The registered tests

Each test had its predictions written and hashed (SHA-256) before the calculation, in the author's working files; the hashes are not publicly time-stamped. The original registrations, in Italian, are in `documenti/registrations_original/` with the recorded hashes; `documenti/REGISTERED_TESTS.md` gives abridged English translations and every outcome.

Counting unit: the clauses tabulated separately in REGISTERED_TESTS.md (P1, P4, P5 and P11c are each split into two clauses; two arithmetic items are not counted). A prediction that passed only in a check made afterwards is counted as failed. A prediction registered for two runs, of which one did not converge, is counted as incomplete.

tests	subject	passed in full / failed / incomplete
P1-P7	the clock	10 / 3 / 0
P8-P9	the checkerboard in 1D and 3D	10 / 0 / 0
P10-P11	mass, relations among particles	6 / 3 / 0
P12-P15	the hand on the boundary	8 / 4 / 0
P16-P19	the field of MM 2.27	8 / 5 / 0
P20	quantized field and matter, 1+1	2 / 1 / 0
P21-P22	the atom	5 / 4 / 0
P23-P25c	the moving atomic clock	10 / 5 / 6
all		**59 / 25 / 6**

The six incomplete are the four predictions of P25 and P23a-b, which passed for the heavier nucleus and were not concluded for the lighter one. Counting instead the items as originally labelled, without the splits, gives 56 / 24 / 6. These totals describe a heterogeneous set of checks; they are not a statistical measure of the model.

Failed predictions and their reading:

- P4 (and its hand estimate) and P5 at $v = 0.8$: thresholds too tight, and an estimate that ignored that the longest chain picks the best path (P5: about 780 copies; the equal-proper-time comparison passed). The overall criterion for P1-P5 was therefore not met (section 3);
- P10b at $\theta = 0.01$: derivative step too large; P10d: the excess of inertial over rest mass in 3D is about $4\theta^2/3$, not below θ^2;
- P11c: the prediction was 8 Weyl nodes, all with chirality +1 (net +4 at $\omega = 0$); there are 16 with net chirality 0 (the prediction used the wrong rotation condition);
- P12a-c: the detector failed because the grid contained $k = 0$, where the two boundary modes are degenerate (all three passed only in a recheck with a shifted grid); P13a: the decay is not exactly exponential (R^2 0.996 against 0.999);
- P16c: the electric-field velocity oscillated at switch-on (Zitterbewegung), read as a delay; P17b-c: packet tails, and a torus too small; P18c: the projector $V V\dagger$ used was wrong with degenerate eigenvectors (the corrected projector keeps the neutral packets at 12 within 1.5×10^{-5} at $t = 20$ and 0.019 at $t = 30$, still above the registered 10^{-6}); P19b: the black-body deviation changes sign later than predicted;
- P20c: Schwinger mass not reached at $N \leq 20$;
- P21a-d: the small atom;
- P24b, P24d, P25b-b, P25b-c, P25b-d: no slowdown with the lattice dispersions.

The addition of version 2.40.2 (section 3.1) has its own registered tests (`aggiunta_2.40.2/REGISTERED_TESTS_2402.md`): 20 passed, 4 failed, 4 not evaluable or incomplete. The totals above are unchanged.

12. What is not claimed

- That any structure in this deposit is new: the time dilation of random events is causal-set theory, the checkerboard and its 3D form are known Dirac quantum walks, the boundary fermions, the lattice field, the Schwinger model and the atom on a lattice are textbook lattice physics. What is proposed is a sequence motivated by the premises of section 2, with the further assumptions stated there.
- That the value of any mass is derived: θ is free, and only relations among masses follow.
- That the checkerboard reproduces the moving atomic clock: it does not at the spacing computed here; P25c, with continuum dispersions, supports the role of lattice kinematics but is not a test of the checkerboard's interacting continuum limit (section 10).
- That the Breit operator used is derived from the lattice field: it is the regularized continuum form.
- That quantum electrodynamics is solved: in 3D the photons enter through leading-order emission rates and a first-order Breit shift; in 1+1 finite chains are diagonalized exactly and the continuum limit is not reached (P20c).
- That the zero-point energy of the field explains the dark energy: with a Planck step it is about 10^{123} times too large.
- That the lattice step is derived: it is a scale of the model; the Planck value is an assumption.
- A curved spacetime: everything here is in flat spacetime.
- That the number of spatial dimensions is derived. Sections 3-10 take three spatial directions as given. With the pairwise rule of section 3.1, n counted sides give $n - 1$ spatial directions, so three directions correspond to four sides. The number of sides is an input, chosen to match the observed 3+1 spacetime: the framework admits three dimensions, it does not derive them (added in 2.40.2).

13. Relation to prior work

element of this deposit	prior work	what differs here
time dilation from random events (section 3)	Brightwell and Gregory (1991); Bombelli, Henson and Sorkin (2009); Ulam's problem (Hammersley 1972; Vershik and Kerov 1977; Logan and Shepp 1977), with the finite-size correction of Baik, Deift and Johansson (1999); Bondi's k-calculus (1964)	premise A motivates the randomness (independence and uniformity are assumed), and B and C the product rule

proper time as the geometric mean of two counts (section 3)	Knuth and Bahreyni (2014): an observer is a chain of an ordered set of events; an interval is quantified by its projections Δp, Δq on two chains, and the consistent scalar is $\sqrt{(\Delta p\,\Delta q)}$, with $\Delta p\,\Delta q = ((\Delta p + \Delta q)/2)^2 - ((\Delta p - \Delta q)/2)^2$; the Minkowski metric and the Lorentz transformations follow (added in 2.40.2)	the pair is a pair of coprime light sides (premise B) and the copies are random (premise A); the identity between geometric and arithmetic mean is the same
more than two sides (section 3.1)	real null frames, used by Zeeman (1964) in the proof that causality implies the Lorentz group; symmetric ones in four dimensions: Derrick (1981), Coll and Morales (1991); in emission (GPS) coordinates it is the coordinate covectors that are null: Rovelli (2002), Coll and Pozo (2006). Knuth (2016, abstract of a talk; full text not consulted) reports that with coordinated chains a consistent quantification exists only for two chains (1+1) or four (3+1)	the sides are independent random counts, and the pairwise rule is tested against the product rule on growing structures; with this rule three sides also give a 2+1 structure, so the setting differs from Knuth's, whose selection of four chains is not addressed here
the checkerboard, 1D (section 4)	Feynman and Hibbs (1965); Jacobson and Schulman (1984); Gaveau, Jacobson, Kac and Schulman (1984); Kac (1974); Bisio, D'Ariano and Tosini (2015)	the turns are changes between the two light sides of the clock of section 3
the 3D Dirac quantum walk, spin ½ (section 4)	Jacobson (1984); Bialynicki-Birula (1994); Meyer (1996); Arrighi, Nesme and Forets (2014); D'Ariano and Perinotti (2014)	a split-step palindromic construction in that family; D'Ariano and Perinotti derive the walk from principles of information processing, here it continues the counting clock
doubling, net chirality zero (section 5)	Nielsen and Ninomiya (1981); Wilson (1977)	a discrete-time automaton, where the theorem's hypotheses are not those of a static Hamiltonian
boundary fermions and their mass (section 6)	Jackiw and Rebbi (1976), for the localization; Kaplan (1992); Shamir (1993), for the chiral lattice construction	—
gauge field and Maxwell on the lattice (section 7)	Peierls (1933); Yee (1966); Wilson (1974); the equations of MM 2.27	the field is the one of MM 2.27, coupled to the counting clock
Casimir, black body (section 7)	Casimir (1948); Planck (1900)	—
Schwinger model (section 8)	Schwinger (1962); Kogut and Susskind (1975); Banks, Kogut and Susskind (1976); Coleman (1976)	—
spontaneous emission (section 9)	the golden rule (textbook form in Fermi 1950); Bethe and Salpeter (1957)	—
moving bound states, Breit (section 10)	Breit (1929); Darwin (1920); Krajcik and Foldy (1974); no-pair projection: Brown and Ravenhall (1951), Sucher (1980); SLAC derivative: Drell, Weinstein and Yankielowicz (1976)	the lattice diagnosis: the slowdown is missing at $M{\cdot}a \approx 2$ and nearly present with continuum dispersions

The literature check was made by the author's AI assistant on the works listed here, not on the whole literature; the bibliographic data were not all checked against the primary sources.

14. How to reproduce (Python 3 with numpy and scipy)

```
python codice/run_all.py        # every program that runs on a laptop (8 GB), about one hour;
                                # outputs in uscite/, compared with risultati/
python codice/tabelle_aws.py    # recomputes the tables of sections 9-10 from the reference outputs
in risultati/aws/
```

`run_all.py` copies the programs into `uscite/` and runs them there, because some rechecks read the main program by its file name. Running times printed by the programs are masked in the comparison; JSON files are compared by content, with numbers equal within 10^{-9} relative (the eigensolvers differ in the last digits from run to run). `tabelle_aws.py` rebuilds the tables; it does not re-check every registered condition (for P22 it prints one state per multiplet).

The atoms of sections 9-10 need about 40 GB of memory per run and were computed on AWS (r7i.4xlarge, 16 vCPU, 128 GB; 25-160 minutes per run). Their programs are in `codice/`, their outputs in `risultati/aws/`, and the commands are:

```
python atomo_grande.py 144 0.625 0.2 0.3 <folder>                    # P22
```

```
python atomo_nucleo.py 144 0.625 0.2 0.375 2.2222 <V> <folder>            # P23 (V = 0, 0.15, 0.30)
python breit.py <folder> 144 0.625 0.2 0.375 2.2222 <V>                   # P24
python atomo_nucleo_np.py 144 0.625 <r> 0.375 2.2222 <V> <folder>         # P25b (r = 0.2, 0.1,
0.05)
python atomo_nucleo_slac.py 144 0.625 0 0.375 2.2222 <V> <folder>         # P25c (r must be 0)
python breit_slac.py <folder> 144 0.625 0 0.375 2.2222 <V>                # P25c
```

The environment variable MM_WORKERS sets the number of FFT threads. In `atomo_nucleo_slac.py` the Wilson argument must be 0: a non-zero value would change the reduced mass printed in the diagnostics without adding a Wilson term. Outputs of the tests made before registration (torus 32^3-64^3) are in `risultati/prove/`.

The addition of 2.40.2 (section 3.1) has its own programs in `aggiunta_2.40.2/codice/` (they also need sympy); the commands, about 80 minutes in all on two processes, are in `aggiunta_2.40.2/ADDITION.md`, section 7, and `aggiunta_2.40.2/codice/collaudo_2402.py` repeats the check made before deposit.

SHA-256 fingerprints are in `IMPRONTE.sha256`. Code comments and printed labels are in Italian.

15. Glossary

term	meaning here
copy	an event of the count (premise A)
light sides a, b	the two sides of the light diamond between the start and the end of a clock
turn	a change of light direction along a history; its weight carries the mass
checkerboard	the sum over histories with an imaginary weight per turn; in 3D, a Dirac quantum walk
Wilson term r	a turn that depends on the neighbours; it makes the extra lattice species heavy
hand	chirality; a one-handed state is a Weyl fermion
boundary	a surface where the turn changes sign
R	$\omega(P)/\omega(0)$: the n = 2 centroid minus 1s gap of the moving atom over its value at rest
V	nominal velocity $P/\sqrt{((m + M)^2 + P^2)}$
no-pair	the projection on positive-energy states of the free electron
sides $x_1, \ldots, x_n$	independent counts of a clock; for n = 2, the light sides a, b (section 3.1)
pairwise rule	$Q = \Sigma_{i<j}\, x_i x_j$: each pair of sides multiplies and the pairs add; the clock reads $\sqrt{Q}$

16. Provenance

The author is not a professional physicist. The framework and its premises are his. The computations were done with an AI assistant (Claude). The predictions of the numbered tests were hashed before computing in the author's working files; these hashes are not publicly time-stamped. Checks made afterwards are labelled as such. A second AI assistant (Codex) reviewed the text and the code of the moving-atom programs, with independent analytical checks (the continuum coefficients, the negative-energy band, the no-pair operator, the counting of the predictions), without executing the programs or consulting external sources; its corrections are applied. For version 2.40.2 the same reviewer (Codex, model gpt-6-luna) read the addition in two rounds, without executing the programs; its corrections are applied. No human peer review is claimed. Before deposit the author's assistant re-ran from scratch every program that fits on a laptop and compared the outputs with the reference files. For the addition of 2.40.2 the check re-ran the algebra, the clock ratios, the containment diagnostic and the 3+1 control, re-analysed the saved outputs of the rank structures and re-ran one structure and one moving-clock run of them (`aggiunta_2.40.2/risultati/COLLAUDO.txt`); the full set of those runs takes about 70 minutes on two processes and was not repeated.

17. References

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Sintesi in italiano

- **Il tempo dal conteggio (P1-P7).** Con l'ordine dei coni di luce e un orologio che legge la catena più lunga, eventi a caso danno il tempo proprio di Einstein; su una griglia la catena non dipende dalla velocità. Un orologio che conta gli eventi e moltiplica le due direzioni di luce ridà il fattore di Lorentz anche sulla griglia: la media geometrica dei due lati contro quella aritmetica. Il criterio complessivo registrato per P1-P5 non è stato soddisfatto (P5 a v = 0,8, per la taglia).
- **Più di due lati (aggiunta nella 2.40.2).** Se un orologio ha n conteggi (lati), la regola del prodotto di tutti i lati non ha Lorentz da tre lati in su (con quattro lati gli orologi in moto sbagliano del 12 %); la regola delle coppie (ogni coppia si moltiplica, le coppie si sommano) sì: n lati danno n − 1 direzioni dello spazio, nelle quali la forma è isotropa, e con quattro lati il 3+1 (dimensione 3,92 sui sei intervalli stimati dentro la regione almeno al 95 %; il confronto registrato fra due osservatori non è superato). L'algebra è nota (coordinate lungo direzioni di luce con prodotti uguali); la prova dice quale estensione regge sulle strutture contate.
- **La scacchiera (P8-P9).** Con un peso immaginario per ogni svolta lo stesso conteggio è la scacchiera di Feynman, un cammino quantistico di Dirac; nella costruzione tridimensionale usata qui lo spin ½ è necessario.
- **La massa (P10-P11).** La massa è l'angolo di svolta per passo; nell'automa in una dimensione la banda diventa piatta a $\theta = \pi/2$; particella e antiparticella hanno la stessa massa; le particelle leggere cadono allo stesso modo; le mani vengono in coppia (16 nodi di Weyl con chiralità netta zero; il palindromo con massa ha 64 valli di Dirac, corretto nella 2.40.1). Il valore della singola massa non si ricava.
- **La mano sul bordo (P12-P15).** Sul confine fra due zone di svolta opposta vive una particella senza massa con una mano sola; due confini si legano in una massa che cala con la distanza. Con i parametri usati ci sono anche stati di parete vicino a metà periodo (gemelle nel tempo); nel controllo con $\theta_0 = 0{,}1 < r$ non ci sono (aggiunto nella 2.40.1).
- **Il campo della 2.27 (P16-P20).** Invarianza di gauge, livelli di Landau, equazioni di evoluzione di Maxwell in 2D e 3D con la legge di Coulomb (la legge di Gauss non è esatta, come nella 2.27), fotoni con due polarizzazioni, coefficiente dell'energia di Casimir allo 0,03 %; in una dimensione confinamento e rottura della corda.
- **L'atomo (P21-P22).** Un atomo con raggio di Bohr di sei passi emette luce con i ritmi della formula idrogenoide (stessa $Z\alpha$ e massa efficace) entro l'1,7 %; l'atomo piccolo non è quantitativo.
- **L'orologio atomico in moto (P23-P25c).** Sul reticolo calcolabile (nucleo più pesante di un'unità di reticolo) l'atomo in moto non rallenta; con le dispersioni del continuo il coefficiente è −0,44 (−0,49 al primo ordine stretto) contro −0,50. Questo indica che la cinematica del reticolo pesa molto, ma non è una prova del limite continuo della scacchiera con interazione: l'operatore di Breit usato è la forma del continuo, non ricavata dal reticolo.
- **Conteggio:** 59 previsioni superate per intero, 25 non superate, 6 incomplete (le 4 di P25 e due di P23 riuscite per un solo nucleo); nell'aggiunta 2.40.2: 20 superate, 4 non superate, 4 non valutabili o incomplete.
- **Che cosa non si afferma:** che le strutture siano nuove (sono tutte note); che il valore delle masse sia ricavato; che la scacchiera, alla grana calcolata, dia l'orologio in moto; che il numero delle dimensioni sia ricavato (MM ammette le tre dimensioni: il numero dei lati è un ingresso, scelto per accordarsi con il 3+1 osservato).
- **Precedenti aggiunti nella 2.40.2:** Knuth e Bahreyni (2014) ricavano il tempo proprio come $\sqrt{(\Delta p \cdot \Delta q)}$ da proiezioni su due catene; secondo l'abstract di un intervento del 2016, Knuth trova che catene coordinate danno una misura coerente solo con due o quattro catene.
- **Provenienza:** l'autore non è un fisico di professione; i calcoli sono fatti con un assistente IA e il testo è stato rivisto da un secondo assistente (Codex) senza rieseguire i programmi (per l'aggiunta 2.40.2 in due giri); le previsioni sono registrate prima dei conti, anche quelle non superate.